

Element B

Jake Steinberg, Anne-Marie Georgi, Astrid Santiago, Jack Hanson, Caleb Bridgeman

After conducting in-depth research on multiple MIP devices, our group now has a greater understanding of the principles we need to consider when designing. Our group researched five different MIP devices and recorded how they work, their price, and their pros and cons.

Attempts and Analysis of Attempts:

1. POWERBREATHE K-Series IMT

The green and white device is an older version of the MIP Respiratory device. The green POWERBREATHE K-Series IMT is \$525.00. The orange and black device is a newer version. This is \$750.00.



Summary:

The K-series IMT is a group of medical devices that serve the same purpose of breathing training for individuals. The website says, “It achieves this by automatically adapting to increases in your [inspiratory](#) muscle strength at the beginning of each training session. What this means is that the training load is highest at the start of your inhalation and then gradually decreases as your lungs fill with air.” The Powerbreathe K1 is a hand-held, electronic inspiratory muscle training device offering entry-level breathing training without software. The K1 only stores data from your last

training session. The K-5 series is the newest, most advanced model onboard memory. This one stores up to 36 sessions of the most important biofeedback marker. It can connect the advanced Breathe-Link Live feedback software. You have a sophisticated testing, training, and monitoring breathing training system.

Helpful Pieces:

A helpful aspect of these devices (Including the early devices without software) is that they can store data from multiple sessions.

Unhelpful Pieces:

The devices range from \$525.00 to \$750.00, meaning many people/ companies want to buy them in bulk or will find a cheaper option.

Source:

<https://www.powerbreathe.com/us/product-category/breathing-trainers-imt-and-emt/k-series-imt/>

2. Philips Threshold IMT

Summary:

The Philips Threshold IMT is a respiratory device that helps patients strengthen their breathing muscles, such as their diaphragm. It is designed to adjust the resistance of airflow (PHILIPS).

The device is also portable and lightweight, allowing users to take it wherever they like. The threshold loading mechanism has a one-way, spring-loaded valve, allowing constant resistance no matter the user's breathing speed. Due to these



features, it is a common device among patients needing respiratory strength and endurance.

Helpful Pieces

One helpful feature of the Philips Threshold IMT is its customizability regarding resistance. This allows users to adjust their resistance over time, starting at a comfortable level and gradually increasing it.

To continue, the Philips Threshold IMT is exceptionally durable and simple. Fewer parts can break, making this device more cost-effective for long-term use.

Customer feedback and clinical doctors say the device is reliable and increases respiratory muscle endurance.

Unhelpful Pieces

However, although the Philips Threshold IMT has many helpful parts, it also has a few unhelpful parts. For example, many users say the device's instructions are inadequate. This isn't good for beginners trying to start their training in respiratory endurance.

Unlike the POWERbreathe K-Series IMT, the Philips Threshold IMT doesn't have a feedback mechanism. This isn't good because it could discourage users from training as they might not see any progress. However, this shouldn't be a big surprise as the device is cheaper than other respiratory devices.

Lastly, the device only focuses on inspiratory muscles rather than both inspiratory and expiratory muscles. This could be a drawback for users who need in-depth respiratory muscle training. As

mentioned before, this isn't a significant issue, as the device is about \$200.00, so it won't be the most advanced device on the market.

https://www.bostonorthoresp.com/Philips-Threshold-IMT_p_9630.html

<https://www.documents.philips.com/assets/20200609/5a5dd0ba4d7149eebf55abd50033347d.pdf>

3. HealthScan PFlex Trainer

The objective of this device is to exercise the diaphragm.

This highly effective inspiratory muscle trainer is designed to increase respiratory muscle strength. "Improved inspiratory muscle strength and endurance could potentially improve cough and maximal exercise ventilation in addition to decreasing dyspnea. "The inspiratory muscles can be trained

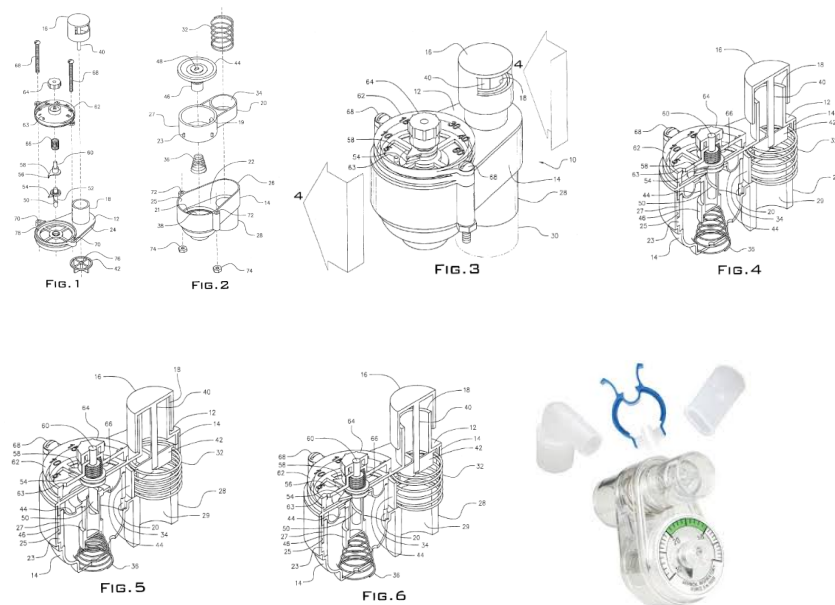


like the limb muscles with inexpensive devices that increase the resistive or threshold inspiratory load on the inspiratory muscles" (Reid et al. 2010). The way to use this is to put the mouthpiece in the mouth. The larger side is the mouthpiece. Then, inhale to work out the diaphragm. The circular component on the top of the device rotates, changing the inhaling resistance. The principle is similar to adding weights for lifting weights to achieve stronger muscles.

Source

<https://scireproject.com/evidence/respiratory-management-rehab-phase/respiratory-muscle-training/>

Helpful Pieces:



The patent describes a disposable manometer device developed for measuring a patient's maximal inspiratory pressure, specifically aiding decisions on whether a patient can be weaned off mechanical ventilation. This device is single-use and compact, providing a method to gauge the negative pressure (suction) generated when a patient inhales. The device features a patient port that connects to the patient's airway and an entrance port to allow ambient air entry. The device creates a vacuum by briefly blocking this port, recording the maximum negative air pressure a patient can produce during inhalation. A key feature is the inclusion of a spring-loaded diaphragm and pointers within the manometer, which react to the inhaled air's suction force and display a reading of the patient's inspiratory strength. The device design enables healthcare providers to monitor and record inspiratory pressures easily, using visual indicators to capture peak effort. This method ensures that patient respiratory strength is measured directly, supporting more precise care decisions without relying on complex or costly equipment.

Helpful Pieces

The spring's concept as a measuring device is solid. A spring allows the device to be calibrated to different inhalation strengths. A person inhales, pulling the tab against the spring, pushing back, and providing pressure. An electronic device or mechanical system can be used to measure the pressure. The device's compactness is also good, as it can be moved and passed around quickly. The mouthpiece is a solid baseline, though we may have to change it slightly. This will allow the device to be reused, with each person having a personal mouthpiece or enough mouthpieces to be swapped and cleaned as the other is used.

Unhelpful Pieces

Single-use is unhelpful because the device we build must be able to be used by many students in a short period of time. The dial measurement will also have to go. If we can change the shape of

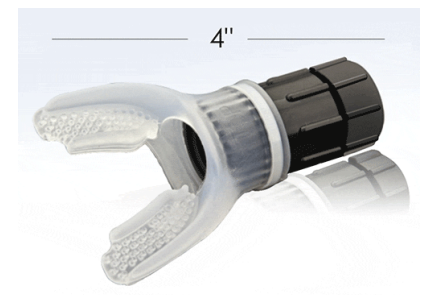
the device and use another measurement recorder, the device can be streamlined and improved to be more user-friendly.

[NIFometer](#)

5. Expand-A-Lung

The Expand-A-Lung Breathing Fitness Trainer is a respiratory training device that enhances lung strength and endurance by providing adjustable inspiratory and expiratory resistance. It benefits athletes and individuals with respiratory issues like COPD by enabling deeper and more efficient breathing. Users can use the device to improve respiratory muscle strength, enhance oxygen intake, reduce

breathlessness, and improve stamina during physical activities. Studies support that respiratory training with devices like Expand-A-Lung can significantly improve endurance by strengthening the diaphragm and intercostal muscles, which are essential for better performance and reduced muscle fatigue. "Selected the #1 breathing fitness trainer by the L.A Times and featured in the following sports magazines: runner's world • triathlete • australian triathlon & multisport • velonews • international freediving • hawaii skin diver • freesurf hawaii • the box crossfit • black belt martial arts • international kickboxing"



Helpful pieces

Flexible Silicone mouthpiece provides comfort and the best fit and seal.

The entire device is pocket-sized and can be taken anywhere.

Heavy-duty plastic which provides durability.

Wide range of resistance levels.

Easy to use and has only one moving part. Turn the valve to the left to decrease the air flow in order to create more resistance. Turn it to the right for the opposite.

Unhelpful Pieces

It benefits many users, but without medical supervision, the device might not be suitable for individuals with certain severe respiratory conditions.

Adjustable but may not be precise enough for users with specific respiratory needs

<https://trello.com/invite/b/6736296627068320972e3045/ATTIc27839eaec53a059dcd5497d8155278e52AAF2D7/drexel-project-2024>